Summary

- The area of the region bounded by the curve $y = f(x)$ between limits $x = a$ *and* $x = b$ with x –axis if area lies above x-axis is $= \int_a^b y\,dx$.
- The area of the region bounded by the curve $y = f(x)$ between the limits $x = a$ and $x = b$ with $x - axis$ if area lies below $x - axis$ is $= \int_a^b -y dx$
- The area of the region bounded by the curve $x = g(y)$ between te limits $y = c$ and $y = d$ with y - axis if the area lies to the right of y- axis is $= \int_c^d x\ dy$
- The area of the region bounded by the curve $x = g(y)$ between the limits $y=d$ and $y=e$ with y-axis if the area lies to the left of y- axis is $= \int_d^e -x\ dy$
- The area between the two given curves $y=f(x)$ and $y=g(x)$ from $x = a$ *to* $x = b$, is $\int_a^b \left(f(x) - g(x)\right) dx.$
- If the rate of growth or sale of a function is a known function of t say $f(t)$ where t is a time measure, then total growth (or) sale of a product over a time period t is given by,

  Total sale $= \int_0^r f(t)\,dt,\ 0 \le t \le r$
- Elasticity of demand is $\eta_d = \frac{-p}{x}\ \frac{dx}{dp}$
- Total inventory carrying cost $= c_1 \int_0^T I(x)dx$
- Amount of annuity after N Payment is $A = \int_o^N pe^{rt}\,dt$
- Cost function is $C = \int (MC)\,dx + k.$
- Average cost function is $AC = \frac{C}{x}, x \neq 0$
- Revenue function is $R = \int (MR)dx + k.$
- Demand function is $P = \frac{R}{x}$
- Profit function is =MR–MC = $R'(x) - C'(x)$
- Consumer's surplus = $\int_0^{x_0} f(x)\,dx - x_0 p_0$
- Producer's surplus = $x_0 p_0 - \int_0^{x_0} p(x)\,dx$

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